

Student Evaluation System

Report Mid Term Project (B)

Database Systems (CSC-104)

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1 Introduction

The Student Evaluation System is a desktop application built with C# Windows Forms and MySQL. It is designed for a course instructor at UET Lahore to manage the Outcome-Based Education process for a lab course. Teachers currently handle the entire evaluation cycle on paper: they maintain student records by hand, record attendance manually, assign marks against rubrics, and produce result summaries separately. This application brings all of those tasks into one place.

The system stores students, course learning outcomes (CLOs), rubrics with their measurement levels, assessments and their question-level components, attendance sessions, and final evaluation results. When a teacher marks a student's performance, the obtained marks are calculated automatically using the rubric formula defined in the project brief.

1.1 Objectives

- Provide a single interface for managing students, CLOs, rubrics, and assessments.
- Let teachers mark attendance per session and per student in one step.
- Automate mark calculation based on the rubric level assigned to each assessment component.
- Generate assessment-wise and CLO-wise result reports that open in a browser and can be saved as PDF.
- Keep all database communication through parameterised queries to prevent SQL injection.

1.2 Technologies Used

Technology	Purpose
C# Windows Forms	Desktop UI and application events
MySQL 8.0	Relational database backend
MySql.Data (NuGet)	Database connectivity
Visual Studio 2022	Development environment

Table 1: Technologies Used

2 System Architecture

The application uses a three-layer architecture to separate concerns clearly.

Layer	Responsibility
UI (Forms)	Handles user interaction and displays data. Contains no SQL queries and no business rules.
BL (Business Layer)	Validates all input before forwarding to the data layer. Returns human-readable error strings on failure.
DL (Data Layer)	Executes parameterised MySQL queries via a static <code>DatabaseHelper</code> class.

Table 2: Three-Layer Architecture

The `DatabaseHelper` utility class exposes three methods used across all data-layer classes: `ExecuteQuery` for SELECT statements, `ExecuteNonQuery` for INSERT, UPDATE, and DELETE, and `ExecuteScalar` for single-value results such as `LAST_INSERT_ID()`.

3 Database Design

The database is named `projectbdb26` and runs on MySQL 8.0. The schema was provided by the instructor and was not modified.

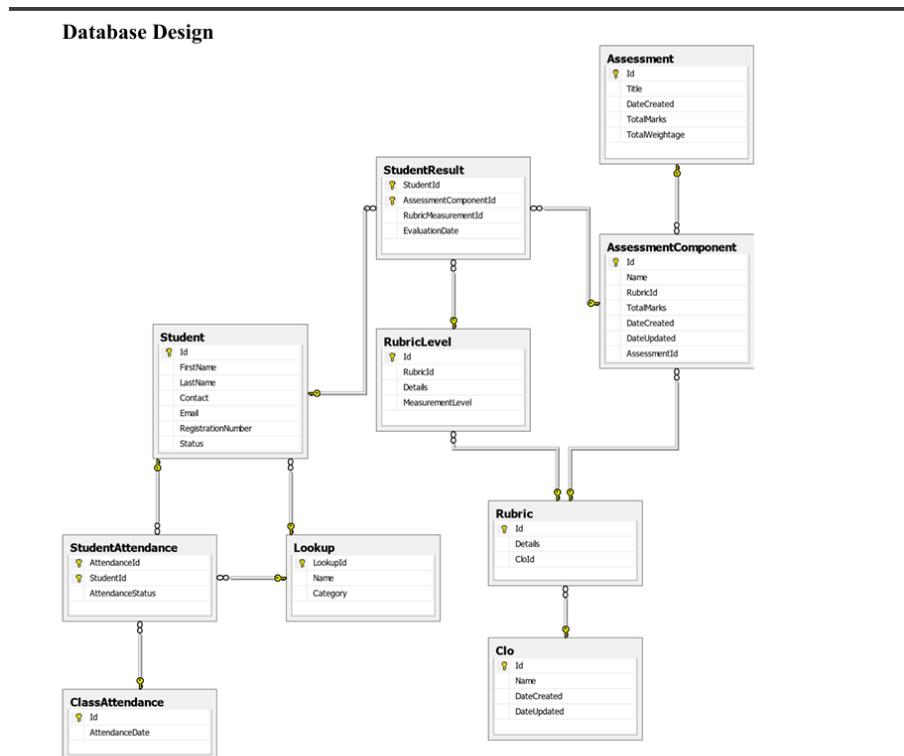


Figure 1: Database Schematic Structure

3.1 Tables Overview

Table	Purpose
student	Stores student personal details and status.
clo	Course Learning Outcomes with creation and update timestamps.
rubric	Each rubric is linked to one CLO and contains a description.
rubriclevel	Each row is one level of a rubric with a numeric measurement level (1 = Unsatisfactory, 4 = Exceptional).
assessment	An assessment (Mid, Lab, Project, etc.) with total marks and weightage.
assessmentcomponent	A question or component inside an assessment, linked to a rubric.
classattendance	One row per class session, stores the date.
studentattendance	Maps a student to a session with a status (Present, Absent, etc.).
studentresult	Stores the rubric level assigned to a student for each component.
lookup	Central reference for status values and attendance codes.

Table 3: Database Tables

3.2 Key Table Structures

Column	Type	Notes
Id	INT	Primary Key, AUTO INCREMENT
FirstName	VARCHAR(50)	NOT NULL
LastName	VARCHAR(50)	Nullable
Contact	VARCHAR(50)	Nullable
Email	VARCHAR(50)	NOT NULL
RegistrationNumber	VARCHAR(20)	NOT NULL
Status	INT	FK to lookup.LookupId

Table 4: Student Table

Column	Type	Notes
Id	INT	Primary Key, AUTO INCREMENT
Name	VARCHAR(50)	NOT NULL
DateCreated	DATETIME(3)	NOT NULL
DateUpdated	DATETIME(3)	NOT NULL

Table 5: CLO Table

Column	Type	Notes
Id	INT	Primary Key (manually assigned)
Details	LONGTEXT	Description of the rubric criterion
CloId	INT	FK to clo.Id

Table 6: Rubric Table

Column	Type	Notes
Id	INT	Primary Key, AUTO INCREMENT
RubricId	INT	FK to rubric.Id
Details	LONGTEXT	Description of the level
MeasurementLevel	INT	Numeric level (1–4)

Table 7: RubricLevel Table

Column	Type	Notes
StudentId	INT	PK + FK to student.Id
AssessmentComponentId	INT	PK + FK to assessmentcomponent.Id
RubricMeasurementId	INT	FK to rubriclevel.Id
EvaluationDate	DATETIME(3)	Timestamp of evaluation

Table 8: StudentResult Table

4 Features and Use Cases

4.1 Main Dashboard

The application opens on a main menu that provides buttons for all six major modules. Each button opens the corresponding form.

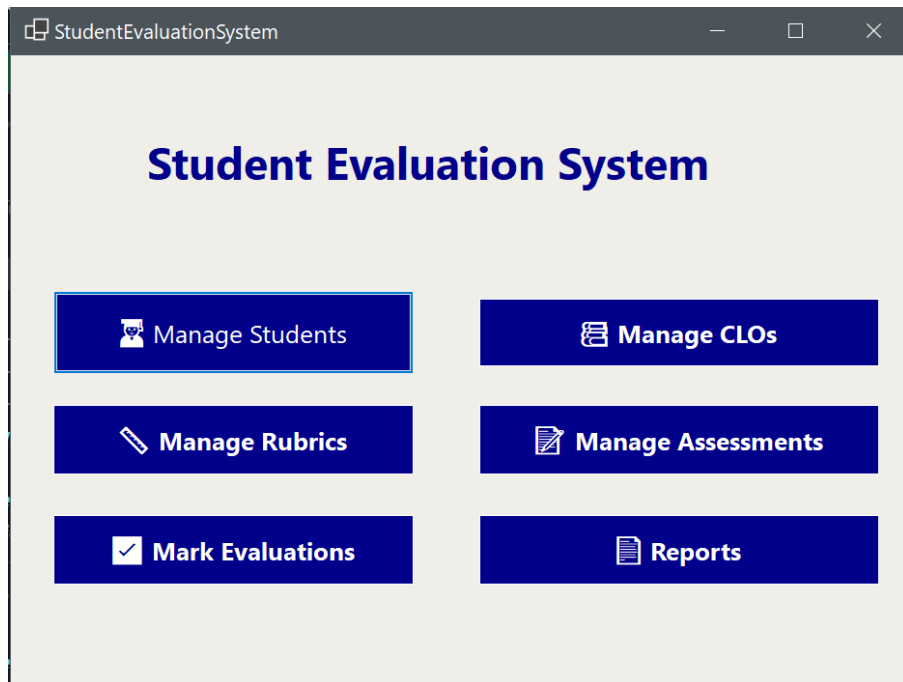


Figure 2: Main Dashboard

4.2 Use Case 1 : Manage Students

Details: The student management menu offers five actions: Add Student, Edit Student, Search Student, Delete Student, and Mark Attendance.

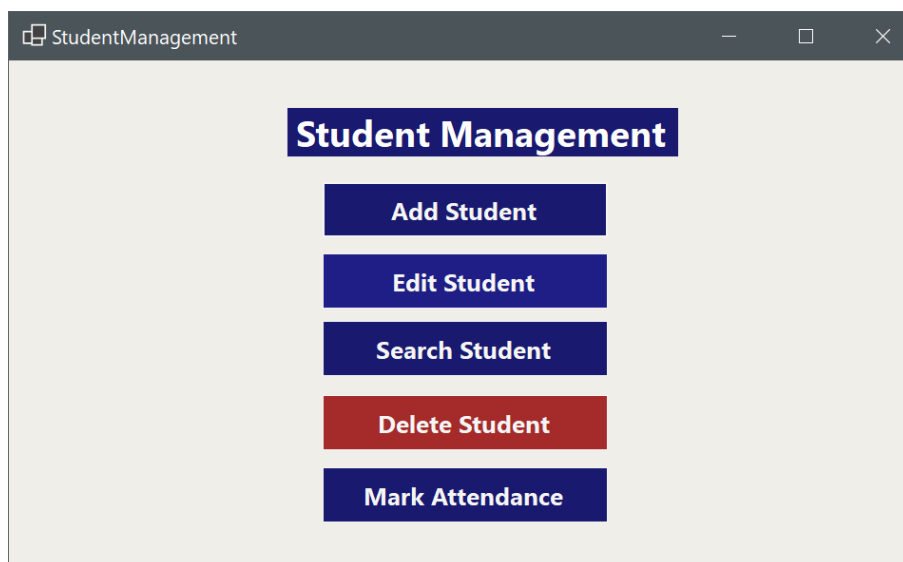


Figure 3: Student Management Menu

4.2.1 Add Student

The Add Student form collects personal information and inserts a new record into the `student` table.

Figure 4: Add New Student Form

Fields and Validations:

Field	Required	Validation
First Name	Yes	Cannot be null or whitespace.
Last Name	No	Optional.
Email	Yes	Cannot be null or whitespace.
Contact	No	Optional.
Registration No	Yes	Cannot be null or whitespace.
Status	Yes	Loaded from <code>lookup</code> table (Active / In-Active).

Table 9: Add Student — Field Validations

Tables used: `student`, `lookup`

4.2.2 Edit Student

The Edit Student form searches for a student by registration number, loads the existing values, and allows the teacher to update them.

Edit Student

2025-CS-60 **Search**

First Name

Last Name

Email

Save Changes

Figure 5: Edit Student Form

Validations: Same as Add Student. First Name and Email are required. If the search returns no result, the fields remain empty and no update is triggered.

Tables used: student

4.2.3 Search Student

Allows the teacher to look up a student by registration number and view their full record.

Search Student

2025-CS-60 **Search**

Field	Value
First Name	Muhammad Umer
Last Name	
Email	theumerfareed@gmail.com
Contact	03420532650
Reg No	2025-CS-60

Figure 6: Search Student Form

Tables used: student

4.2.4 Delete Student

The teacher enters the registration number to locate the student first. Only after clicking Find and confirming is the record deleted. A warning message informs the user the action cannot be undone.

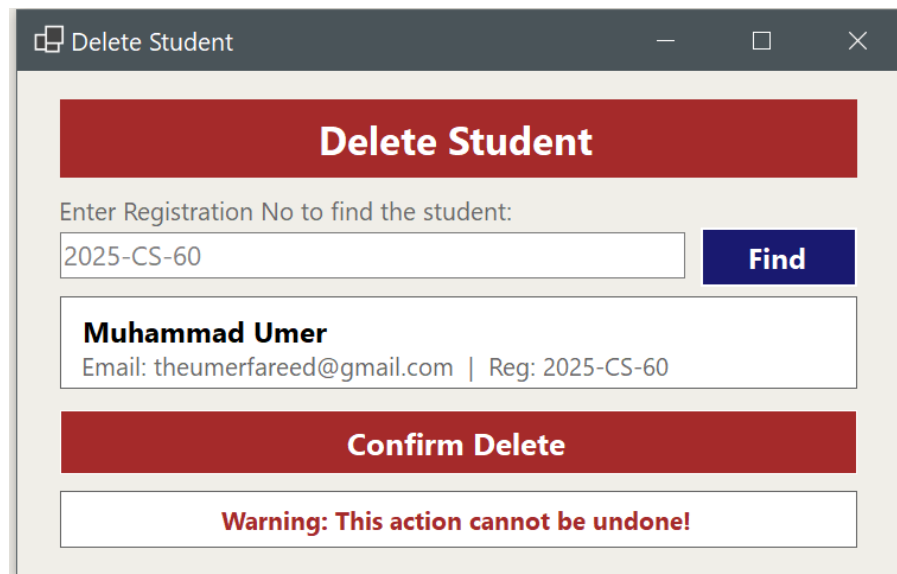


Figure 7: Delete Student Form

Validations: The student must be found before Delete is enabled. If the student has related records (attendance, results), the delete will fail with an appropriate error message.

Tables used: student

4.3 Use Case 2 : Mark Attendance

The teacher selects a class date using a date picker and clicks Load to pull all active students. For each student a dropdown allows setting the status to Present, Absent, Leave, or Late. Clicking Save Attendance commits the session.

Mark Attendance

Class Date: Monday , April 13, 2026  **Load**

Reg No	Name	Status
2025-CS-75	Muhammad Ham...	Present
2025-CS-80	ghuhfh ufeufuhf...	Present
2025-CS-79	bbdhjdejhj fhfhfh...	Present
2025-CS-78	hieihhie3hi hehe...	Present
hdiehdhh	nkdhhidhi idiindh	Present
ibrfhifh	dibabdh kfirif	Present

Save Attendance

Figure 8: Mark Attendance Form

Validations: At least one student must exist. If the student list is empty the save is blocked with a message.

Tables used: classattendance, studentattendance, student, lookup

4.4 Use Case 3 : Manage CLOs

The teacher can view all existing CLOs, add new ones inline, update names, or delete them. Each CLO shows its creation and last-updated timestamp.

#	CLO Name	Action
#30	Window Forms application	Updated: 13 Apr 2026 <button>Delete</button>
#31	Love	Created: 13 Apr 2026 Updated: 13 Apr 2026 <button>Delete</button>
#32	umer	Created: 13 Apr 2026 Updated: 13 Apr 2026 <button>Delete</button>
#34	Window Forms application	Created: 13 Apr 2026 Updated: 13 Apr 2026 <button>Delete</button>
#36	SQL	Created: 13 Apr 2026 Updated: 13 Apr 2026 <button>Delete</button>

[+ add CLO](#)

Save All CLOs

Figure 9: Manage CLOs Form

Validations:

- CLO name cannot be empty.
- Deleting a CLO that is linked to a rubric will fail with an error.

Tables used: clo

4.5 Use Case 4 : Manage Rubrics

The teacher selects a CLO from the dropdown, clicks Load, and sees all rubrics under that CLO. Rubrics can be added (with a manual ID and description) or deleted. A Levels button opens a sub-form to manage the measurement levels for each rubric.

Manage Rubrics

Select a CLO, then manage its rubrics and levels.

Select CLO: SQL Load

ID:	30	DDL	Levels	Del
ID:	101	DML	Levels	Del

[+ add rubric](#)

Save All

Figure 10: Manage Rubrics Form

Levels for: DDL

Rubric Levels (ID 30) — Level 1=Unsatisfactory, 2=Fair, 3=Good, 4=Excellent

Level	4	Extraordinary	▼
Level	3	Good	▼
Level	2	Average	▼
Level	1	Poor	▼

+ Add Save

Figure 11: Manage Rubric Levels Sub-Form

Validations:

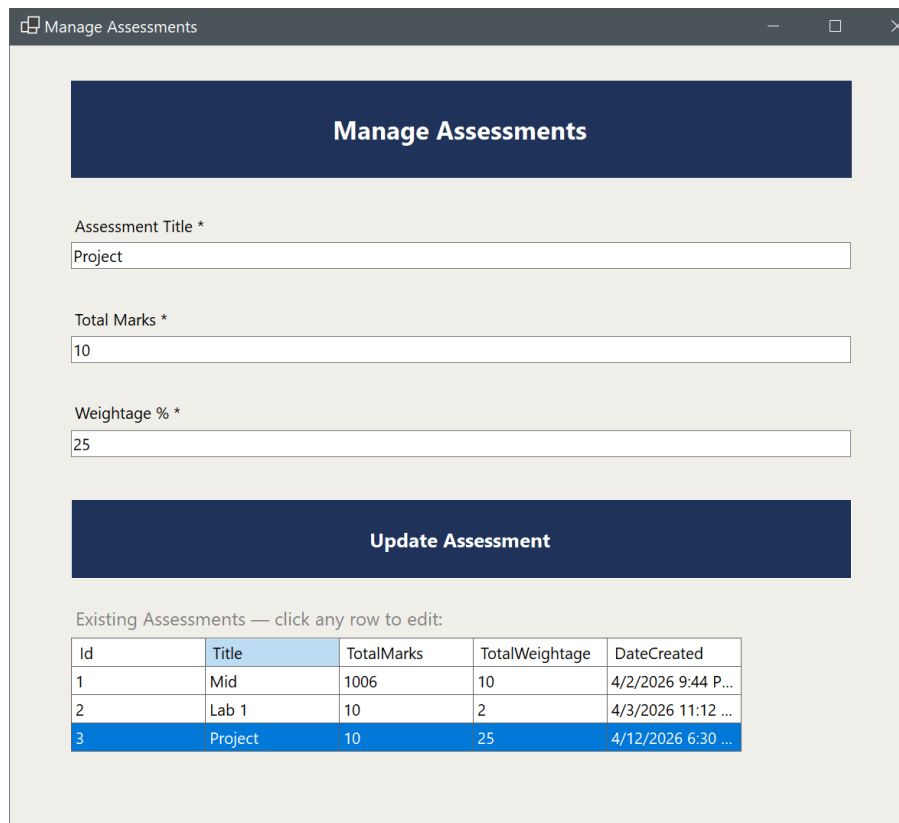
- Rubric ID must be a positive integer.
- Rubric details cannot be empty.
- A CLO must be selected before loading.

- Cannot delete a rubric linked to an assessment component.

Tables used: rubric, rubriclevel, clo

4.6 Use Case 5 : Manage Assessments

The teacher adds assessments (Mid, Lab 1, Project, etc.) with a title, total marks, and weightage percentage. Existing assessments are shown in a grid below and can be clicked to load and edit. A separate Add Questions form lets the teacher attach question components to the assessment, assigning a rubric and marks to each question.



Manage Assessments

Assessment Title *

Project

Total Marks *

10

Weightage % *

25

Update Assessment

Existing Assessments — click any row to edit:

Id	Title	TotalMarks	TotalWeightage	DateCreated
1	Mid	1006	10	4/2/2026 9:44 P ...
2	Lab 1	10	2	4/3/2026 11:12 ...
3	Project	10	25	4/12/2026 6:30 ...

Figure 12: Manage Assessments Form

Add Questions

Assessment: Project Total: 10 | Used: 10 | Remaining: 0

#	Question / Component	Rubric	Marks	Remov
1	UI	Interface	5	X
2	Logic	DML	5	X

[+ add question](#)

Marks used out of —: 10 / 10

Save Questions **Skip for now**

Figure 13: Add Question Component Form

Validations:

- Title is required.
- Total Marks must be greater than 0.
- Weightage must be greater than 0.
- The sum of component marks across all questions cannot exceed the assessment's total marks.
- Cannot delete an assessment that has linked components or student results.

Tables used: assessment, assessmentcomponent, rubric

4.7 Use Case 6 : Mark Evaluations

The teacher selects a student and an assessment. The form loads all components of that assessment and, for each one, shows the rubric name, the maximum level (Max:4), and a dropdown of rubric level descriptions. When a level is chosen, the obtained marks are calculated instantly using the formula:

$$ObtainedMarks = \frac{ObtainedLevel}{MaxLevel} \times ComponentMarks$$

The total obtained marks update at the bottom as each level is selected. Clicking Save Evaluation writes all results to the database.

Mark Evaluation

Select Student *

2025-CS-75 | Muhammad Hamza Hassan

Select Assessment *

Project

Components — assign a level to each:

Component	Rubric	Max	Level	Marks
UI	Interface	Max:4	Excellent	
Logic	DML	Max:4	Good	

Total Obtained Marks: 8.75 / 10

Figure 14: Mark Evaluation Form

Validations:

- A student must be selected.
- An assessment with components must be selected.
- Every component must have a level assigned before saving.

Tables used: studentresult, assessmentcomponent, rubriclevel, student, assessment

4.8 Use Case 7 : Reports

The Reports form provides two report types via a dropdown. Selecting *Assessment Wise Class Result* requires choosing an assessment; selecting *CLO Wise Class Result* generates a result grouped by CLO. Clicking Generate Report builds an HTML report and opens it in the default browser. The teacher can then press Ctrl+P and save as PDF.

Reports

Report Type *

CLO Wise Class Result

Generate PDF Report

* = required fields

Figure 16: (CLO Wise Report Form)

Reports

Report Type *

Assessment Wise Class Result

Select Assessment

Project

Generate PDF Report

* = required fields

Figure 15: Reports Form (Assessment Wise)

Validations:

- An assessment must be selected for the assessment-wise report.
- If no data is found the report is not generated and an info message is shown.

Tables used: studentresult, assessment, assessmentcomponent, rubriclevel, student, clo

5 Reports

5.1 Report 1 : Assessment Wise Class Result

Purpose: Shows every student's obtained marks broken down by component for a selected assessment. Useful for identifying which questions students are struggling with.

Query:

```
SELECT
    s.FirstName,
    s.LastName,
    s.RegistrationNumber,
    ac.Name AS Component,
    rl.MeasurementLevel AS Level,
    ROUND((rl.MeasurementLevel / maxl.MaxLevel) * ac.TotalMarks, 2) AS
        ObtainedMarks,
    ac.TotalMarks AS ComponentMax
FROM studentresult sr
JOIN student s ON sr.StudentId = s.Id
JOIN assessmentcomponent ac ON sr.AssessmentComponentId = ac.Id
JOIN rubriclevel rl ON sr.RubricMeasurementId = rl.Id
JOIN (
    SELECT RubricId, MAX(MeasurementLevel) AS MaxLevel
    FROM rubriclevel
    GROUP BY RubricId
) maxl ON rl.RubricId = maxl.RubricId
WHERE ac.AssessmentId = @AssessmentId
ORDER BY s.RegistrationNumber, ac.Name;
```

Assessment Result — Project

Generated: 13 Apr 2026, 07:30 PM | CS104 Database Systems

RegistrationNumber	StudentName	Component	ComponentMarks	ObtainedLevel	MaxLevel	ObtainedMarks
2025-CS-6		Logic	5	4	4	5.00
2025-CS-6		UI	5	4	4	5.00
2025-CS-60	Muhammad Umer	Logic	5	2	4	2.50
2025-CS-60	Muhammad Umer	UI	5	1	4	1.25
2025-CS-75	Muhammad Hamza Hassan	Logic	5	1	4	1.25
2025-CS-75	Muhammad Hamza Hassan	UI	5	3	4	3.75
2025-CS-78	hieihhie3hi hehehhu	Logic	5	3	4	3.75
2025-CS-78	hieihhie3hi hehehhu	UI	5	2	4	2.50

Figure 17: Assessment Wise Report Output

Business Benefit: The instructor can immediately see which students are below average on a specific component and decide whether to revisit that topic in the next session. The percentage-based calculation ensures fair comparison across components with different max levels.

5.2 Report 2 : CLO Wise Class Result

Purpose: Aggregates each student's performance at the CLO level. Each CLO may map to multiple rubrics, and rubrics appear in multiple assessment components. This report helps the department in its Outcome Based Education review to determine whether a CLO has been achieved by a sufficient proportion of students.

Query:

```
SELECT
    s.RegistrationNumber,
    CONCAT(s.FirstName, ' ', s.LastName) AS StudentName,
    c.Name AS CLO,
    SUM(ROUND((rl.MeasurementLevel / maxl.MaxLevel) * ac.TotalMarks, 2))
        AS ObtainedMarks,
    SUM(ac.TotalMarks) AS TotalMarks
FROM studentresult sr
JOIN student s ON sr.StudentId = s.Id
JOIN assessmentcomponent ac ON sr.AssessmentComponentId = ac.Id
JOIN rubric r ON ac.RubricId = r.Id
JOIN clo c ON r.CloId = c.Id
JOIN rubriclevel rl ON sr.RubricMeasurementId = rl.Id
JOIN (
    SELECT RubricId, MAX(MeasurementLevel) AS MaxLevel
    FROM rubriclevel GROUP BY RubricId
) maxl ON rl.RubricId = maxl.RubricId
GROUP BY s.Id, c.Id
ORDER BY s.RegistrationNumber, c.Name;
```

CLO Wise Class Result

Generated: 13 Apr 2026, 07:30 PM | CS104 Database Systems

CLO	StudentName	RegistrationNumber	Component	ObtainedMarks	ComponentMarks
Love	Muhammad Hamza Hassan	2025-CS-75	Umer	7.50	10
Love	ghuhfh ufeufuhfh	2025-CS-80	Umer	5.00	10
Love	nkdhhdhi idiidhi	hdiehdh	Umer	2.50	10

Figure 18: CLO Report Output

Business Benefit: The department can calculate CLO attainment percentages, flag underperforming CLOs, and document evidence for accreditation bodies such as PEC that course outcomes are being met.

6 Exception Handling and Validations Summary

All user input passes through the Business Layer before reaching the database. Each BL method returns either the string "success" or a plain-language error message that the UI displays in a `MessageBox`. This means the user always sees a readable reason for any failure rather than a raw exception trace.

At the data layer, all queries use `MySQLParameter` objects, so no string concatenation reaches the database, eliminating SQL injection risk.

Key validation rules enforced across the application are listed below.

Entity	Rule
Student	First Name and Email are required. Registration Number is required. Delete fails if the student has attendance or result records.
CLO	Name cannot be empty. Delete fails if rubrics are linked.
Rubric	ID must be positive. Details cannot be empty. A CLO must be selected. Delete fails if the rubric is used in a component.
Assessment	Title required. Marks and weightage must be > 0 . Delete fails if components or results exist.
Attendance	Student list must not be empty before saving.
Mark Evaluation	Student and assessment required. Every component must have a level assigned.
Reports	Assessment required for assessment-wise report. No data produces an info message rather than an empty report.

Table 10: Validation and Error Handling Summary

7 Conclusion

The Student Evaluation System covers the core requirements of the CS104 Database Systems project. It replaces a manual process with a structured desktop application that stores all evaluation data in a normalised MySQL database. The three-layer architecture keeps the codebase maintainable: UI forms contain no SQL, business-layer classes enforce all rules, and data-layer classes handle all database communication through parameterised queries.

The two HTML-based reports provide the instructor with a quick way to review class performance either per assessment or per CLO, which directly supports the Outcome Based Education review process followed by the department.